DynamiClinic

Switching a multi-agent system's architecture per clinical case: what the prototype does, what it shows, and what it does not yet show.



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github.com/LaansDole/multi-agent-simulation
branch threejs-devall

Clinical multi-agent systems commit to *one architecture* before they see the case

MedAgents

Findings of ACL 2024

Multi-round expert consensus on every question: the same heavy pipeline for an easy case and a hard one.

MDAgents

NeurIPS 2024

Picks solo or team once, by complexity. Best on 7 of 10 benchmarks (up to +4.2%), but the structure cannot change mid-case.

Dialectic-Med

Findings of ACL 2026

Proponent, opponent and mediator falsify a hypothesis; fewer hallucinations on medical image QA.

THE PROPOSAL: MAKE THE TOPOLOGY A VARIABLE

Hierarchical — Routine, low-acuity cases. A hub queries specialists, who do not talk to each other.

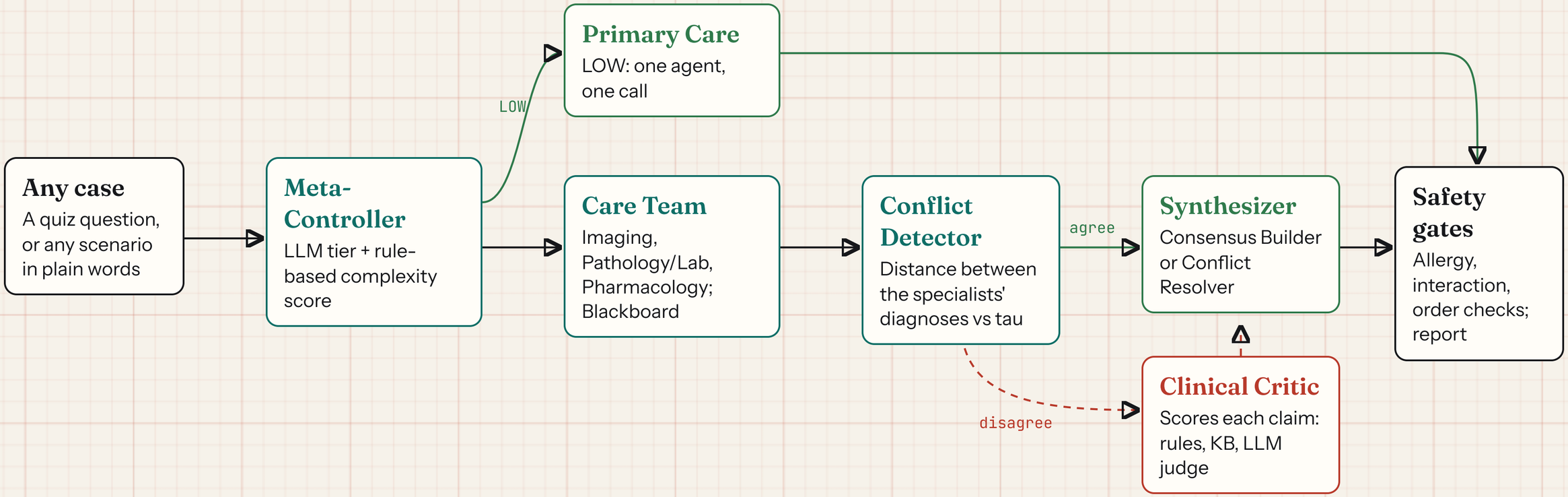
Collaborative — Multi-system cases. Specialists share reasoning and build one differential.

Adversarial — Uncertain or conflicting evidence. A critic attacks the consensus; a mediator resolves.

A **Meta-Controller** scores complexity and uncertainty and switches mode, so heavy compute is spent **only where the case needs it**.

Sources: ACL Anthology 2024.findings-acl.33; arXiv 2404.15155; ACL Anthology 2026.findings-acl.1837

A working pipeline on the DevAll multi-agent platform



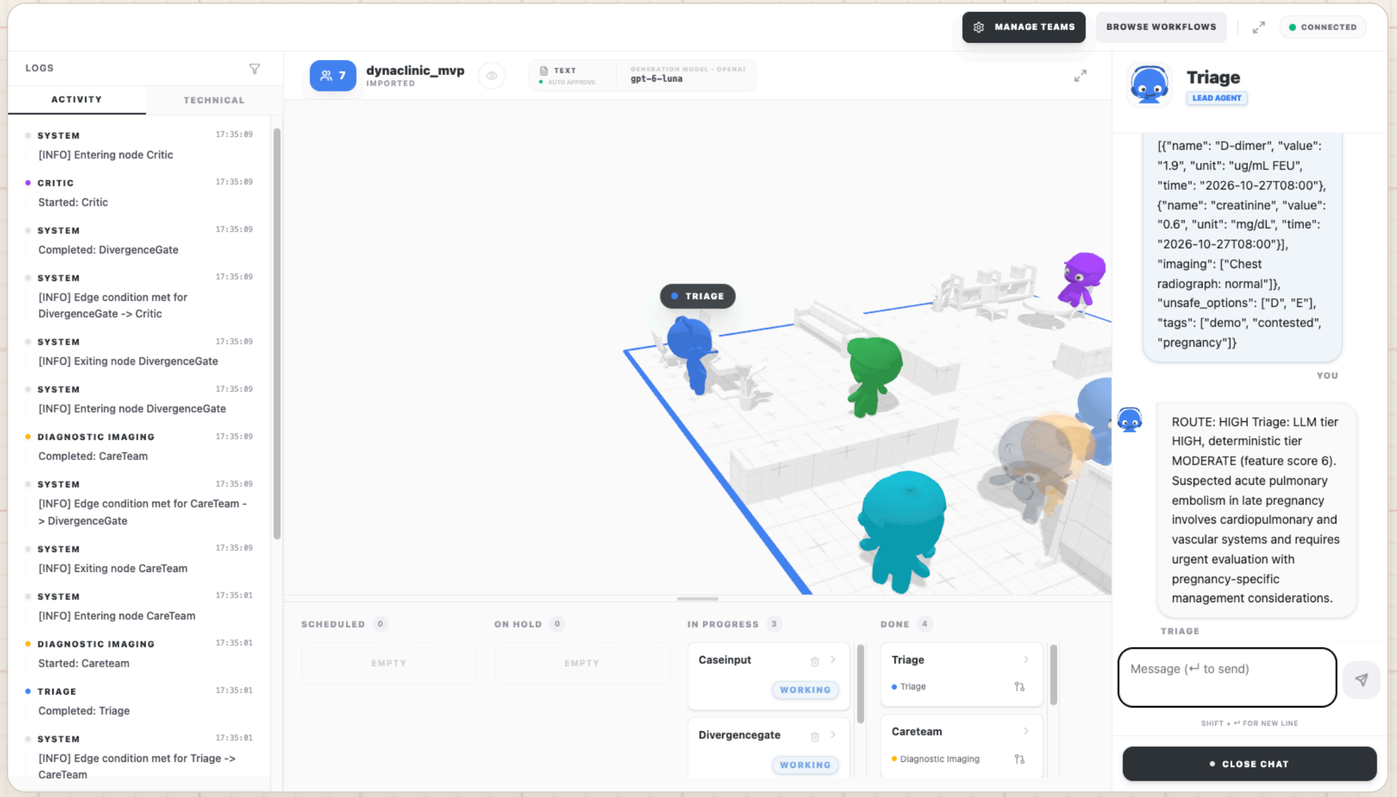
7
custom node types in one YAML workflow

3
arms evaluated: single, dynamic, static (always full team + Critic)

3D
office view: each agent is a character that walks to work

866
automated tests, no model calls needed

The care team disagrees, and the *Critic* is called in



Real run, OpenAI gpt-6-luna. Log: DivergenceGate, then Critic started. Chat: ROUTE: HIGH.

MOCK PATIENT: 30 WEEKS PREGNANT, SUSPECTED CLOT

1. **Triage** calls the case HIGH; all three specialists are consulted.
2. **Imaging** picks B (0.75), **Pharmacology** picks C (0.82), Pathology abstains.
3. **Divergence 0.59 > 0.25**: adversarial mode, the Clinical Critic reviews every claim.
4. The Synthesizer resolves B vs C and answers **C, correct**; safety gates pass.

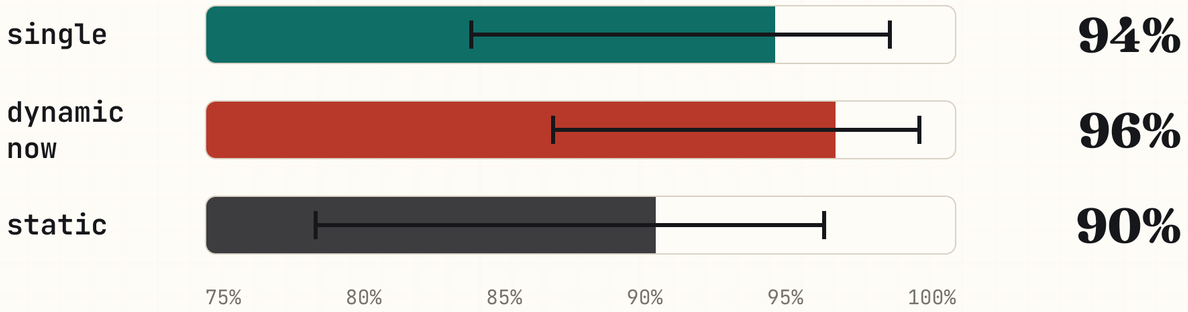
The switch fired in **4 of 9** real runs of this case: all 3 runs triaged HIGH, and 1 of 6 triaged MODERATE. Reported, not forced.

Live demo with video: laansdole.github.io/multi-agent-simulation

The three modes now run. *Not more accurate* than one agent, yet.

MedQA-USMLE, 50 random test questions

Accuracy with 95% interval. gpt-6-luna, one run per arm; "dynamic now" is after f15.



	SINGLE	DYNAMIC NOW	STATIC
MedQA calls / question	1.00	6.94	6.00
MedQA latency, mean	5.3 s	32.0 s	29.9 s
MedQA adversarial mode	n/a	34%	always
Synthetic 24 accuracy	100%	95.8%	100%
Synthetic calls / case	1.00	7.12	6.08
Unsafe answers (synthetic)	0	0	0

Adversarial now runs on 17 of 50 MedQA questions (was 0), but the mesh and debate rounds cost more calls than the old always-on team, and every interval still overlaps. Before f14, dynamic used 36 to 39% fewer calls at 92%.

MedQA: Jin et al. 2021, 4-option test split, seed 2026 sample. Full tables: laansdole.github.io/multi-agent-simulation/results.html

The pipeline and *all three modes* now match; the *thresholds* are not validated

MATCHES

- **Uncertainty score** (entropy of a generalist sweep) and **one rule** over C and U
- Hierarchical hub, **mesh exchange**, **Proponent / Opponent / Mediator**
- Any scenario, not only quiz questions
- Runs on DevAll, as the proposal's Figure 1 shows

PARTIAL

- tau_u calibrated in-sample (p75), not against outcomes
- Conflict Detector threshold not validated
- One round per mode, one agent per role
- H1 / H2 match exact wording, not paraphrases

MISSING

- Counterfactual grounding loss
- Proponent and Opponent **teams**, multi-round debate
- Process models and institutional protocols (L4)

PROPOSAL'S OWN EXAMPLE (FIGURES 2, 3)	EXPECTS	ROUTE, 2 RUNS	MODE NOW (U)	MEDIATOR VERDICT
Sore throat and mild fever for 2 days	LOW	LOW, LOW	adversarial (0.71, 0.65)	H1 acute pharyngitis
68-year-old diabetic, chest pain, renal insufficiency, warfarin	HIGH	HIGH, HIGH	adversarial (0.94, 0.92)	H1 suspected ACS
55-year-old, chest pain, normal ECG, high D-dimer, long flight	CONFLICT	HIGH, HIGH	adversarial (0.91, 0.94)	pulmonary embolism (H1 = H2 reworded)

Details and run folders: laansdole.github.io/multi-agent-simulation/alignment.html

To test the thesis, the benchmark must make the agents *disagree*

1

A harder benchmark

MedXpertQA (ICML 2025): 4,460 expert-level questions in 17 specialties, filtered to be harder than MedQA. Larger samples to narrow the intervals.

2

Headroom for the team

Repeat with a weaker base model, where a single agent is far from the ceiling and specialists can add value.

3

Validate the controller

tau_u sends every proposal example to adversarial: validate it against outcomes, average several generalist sweeps, match paraphrased hypotheses.

4

Safety and validity

Cases with structured data and known unsafe options; thresholds tuned on held-out data; repeated runs; clinician review of the critic's verdicts.

MedXpertQA: arXiv 2501.18362, PMLR v267

What the evidence supports today

1. Switching per case **works end to end** on quiz questions and on any scenario in plain words: triage, team, conflict detection, critic and safety gates run live, and you can watch them in 3D.

2. The proposal's **three modes and one switching rule** now run live; adversarial fires on a third of MedQA, but costs more calls than the old always-on team.

3. It does **not yet beat one strong model**, nor a weak one: with qwen3.5-4b the single agent leads (68% vs 60% dynamic, 62% static). tau_u has to be calibrated per model.

`laansdole.github.io/multi-agent-simulation`
Runbook demo: the full journey on video, step by step

`.../results.html and .../alignment.html`
MedQA and synthetic results; the proposal fact check

`github.com/LaansDole/multi-agent-simulation`
Code, eval harness and data, branch threejs-devall

